



Background Information on Sustainable Agriculture

What is Sustainable Agriculture?

Sustainable agriculture refers to an agricultural production and distribution system that:

- Achieves the integration of natural biological cycles and controls,
- Protects and renews soil fertility and the natural resource base,
- Optimizes the management and use of on-farm resources,
- Reduces the use of non renewable resources and purchased production inputs,
- Provides an adequate and dependable farm income,
- Promotes opportunity in family farming and farm communities, and
- Minimizes adverse impacts on health, safety, wildlife, water quality and the environment.

Perhaps the only constant in our diverse agricultural system is change. Many of today's farmers and ranchers practice a type of agriculture that is far different from how their grandparents, or even their parents, worked the land. Change can be a positive force to improve the world's sources of food and fibre — if those changes are made while balancing profitability, stewardship of natural resources and the health of rural communities.

For generations, farmers have worked to protect and improve the land and the quality of rural life. Early farmers, seeing a need to change some standard agricultural practices, soon became leaders in adopting new farming methods. Some of these new methods were the practice of crop rotation, composting livestock waste and the development of new planting methods. The approaches many of today's farmers and ranchers take as they grapple with tough issues has come to be known as sustainable agriculture.

Sustainable agriculture does not refer to a prescribed set of practices. Instead, it challenges producers to think about the long-term implications of practices and the broad interactions and dynamics of agricultural systems. It also invites consumers to get more involved in agriculture by learning more about and becoming active participants in their food systems. A key goal is to understand agriculture from an ecological perspective — in terms of nutrient and energy dynamics, and interactions among plants, animals, insects and other organisms in agroecosystems — then balance it with profit, community and consumer needs.

Farming methods that improve the sustainability of one farm may not be appropriate to a different farm or region. Each practice must be evaluated in a given farming system for its ability to achieve a set of economic, environmental and social goals. However, we can look to changes adopted by farmers across the country, a few of which are described in these pages, to get a sense of how to improve agricultural sustainability. The farmer case studies that follow demonstrate a few of these: increased crop and landscape diversity, better utilization of on-farm resources such as crop residue and manure, and more effective marketing. These changes and many other alternative approaches are contributing to the goals of lasting farm production, stewardship of land, water and wildlife, and improved quality of life for farmers, their families and rural communities.

<http://www.sarep.ucdavis.edu/concept.htm>

Agriculture has changed dramatically, especially since the end of World War II. Food and fiber productivity soared due to new technologies, mechanization, increased chemical use, specialization and government policies that favored maximizing production. These changes allowed fewer farmers with reduced labor demands to produce more food and fiber.

Although these changes have had many positive effects and reduced many risks in farming, there have also been significant costs. Prominent among these are topsoil depletion, groundwater contamination, the decline of family farms, continued neglect of the living and working conditions for farm laborers, increasing costs of production, and the disintegration of economic and social conditions in rural communities.

A growing movement has emerged during the past two decades to question the role of the agricultural establishment in promoting practices that contribute to these social problems. Today this movement for sustainable agriculture is garnering increasing support and acceptance within mainstream agriculture. Not only does sustainable agriculture address many environmental and social concerns, but it offers innovative and economically viable opportunities for growers, laborers, consumers, policymakers and many others in the entire food system.

Sustainable agriculture integrates three main goals--environmental health, economic profitability, and social and economic equity. A variety of philosophies, policies and practices have contributed to these goals. People in many different capacities, from farmers to consumers, have shared this vision and contributed to it. Despite the diversity of people and perspectives, the following themes commonly weave through definitions of sustainable agriculture.

Sustainability rests on the principle that we must meet the needs of the present without compromising the ability of future generations to meet their own needs. Therefore, *stewardship of both natural and human resources* is of prime importance. Stewardship of human resources includes consideration of social responsibilities such as working and living conditions of laborers, the needs of rural communities, and consumer health and safety both in the present and the future. Stewardship of land and natural resources involves maintaining or enhancing this vital resource base for the long term.

A *systems perspective* is essential to understanding sustainability. The system is envisioned in its broadest sense, from the individual farm, to the local ecosystem, *and* to communities affected by this farming system both locally and globally. An emphasis on the system allows a larger and more thorough view of the consequences of farming practices on both human communities and the environment. A systems approach gives us the tools to explore the interconnections between farming and other aspects of our environment.

A systems approach also implies *interdisciplinary efforts in research and education*. This requires not only the input of researchers from various disciplines, but also farmers, farm-workers, consumers, policymakers and others.

Making the transition to sustainable agriculture is a process. For farmers, the transition to sustainable agriculture normally requires a series of small, realistic steps. Family economics and personal goals influence how fast or how far participants can go in the transition. It is important to realize that each small decision can make a difference and contribute to advancing the entire system further on the "sustainable agriculture continuum." The key to moving forward is the will to take the next step.

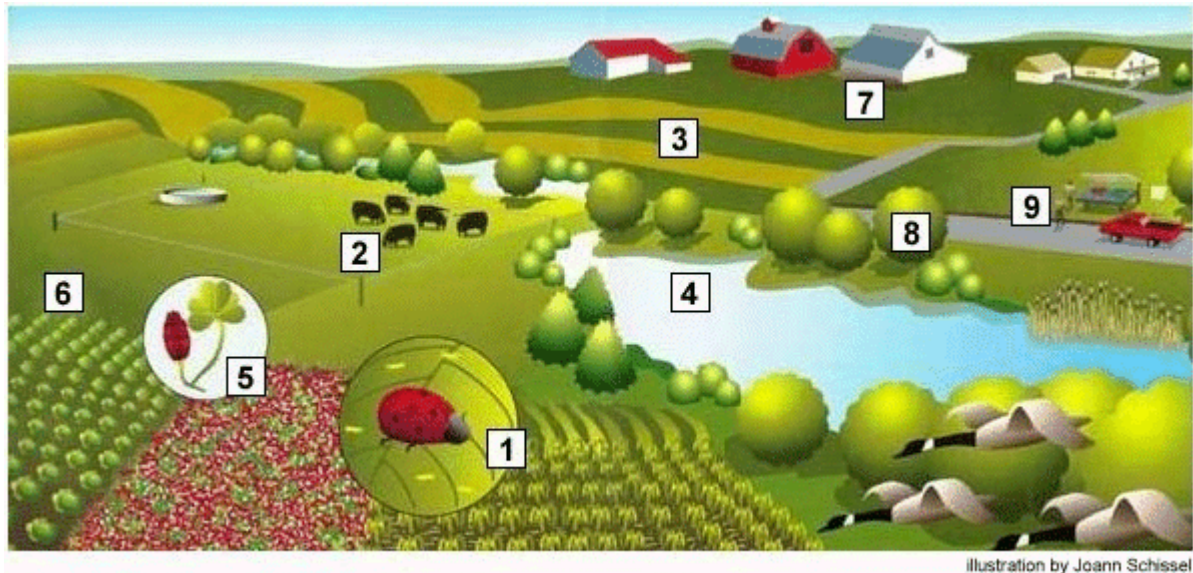
Finally, it is important to point out that *reaching toward the goal of sustainable agriculture is the responsibility of all participants in the system*, including farmers, laborers, policymakers, researchers, retailers, and consumers. Each group has its own part to play, its own unique contribution to make to strengthen the sustainable agriculture community.

The remainder of this document considers specific strategies for realizing these broad themes or goals. The strategies are grouped according to three separate though related areas of concern:

Farming and Natural Resources, Plant and Animal Production Practices, and the Economic, Social and Political Context. They represent a range of potential ideas for individuals committed to interpreting the vision of sustainable agriculture within their own circumstances.

<http://www.sare.org/>

Elements of Sustainability



There are many ways to improve the sustainability of a given farming system, and these vary from region to region. However, there are some common sets of practices among farmers trying to take a more sustainable approach, in part through greater use of on-farm or local resources. Some of those practices are illustrated here, each contributing in some way to long-term farm profitability, environmental stewardship and rural quality of life.

1) Integrated Pest Management (IPM): IPM is a sustainable approach to managing pests by combining biological, cultural, physical and chemical tools in a way that minimizes economic, health and environmental risks.

2) Rotational Grazing: New management-intensive grazing systems take animals out of the barn and into the pasture to provide high-quality forage and reduced feed costs while avoiding manure build up.

3) Soil Conservation: Many soil conservation methods, including strip cropping, reduced tillage and "no-till," help prevent loss of soil due to wind and water erosion.

4) Water Quality/Wetlands: Water conservation and protection have become important parts of agricultural stewardship. Many practices have developed to improve quality of drinking and surface water, as well as to protect wetlands. Wetlands play a key role in filtering nutrients and pesticides, in addition to providing wildlife habitat.

5) Cover Crops: Growing plants such as rye, clover or vetch in the off season after harvesting a grain or vegetable crop can provide several benefits, including weed suppression, erosion control, and improved soil nutrients and soil quality.

6) Crop/Landscape Diversity: Growing a greater variety of crops on a farm can help reduce risks from extremes in weather, market conditions or crop pests. Increased diversity of crops and

other plants, such as trees and shrubs, also can contribute to soil conservation, wildlife habitat and increased populations of beneficial insects.

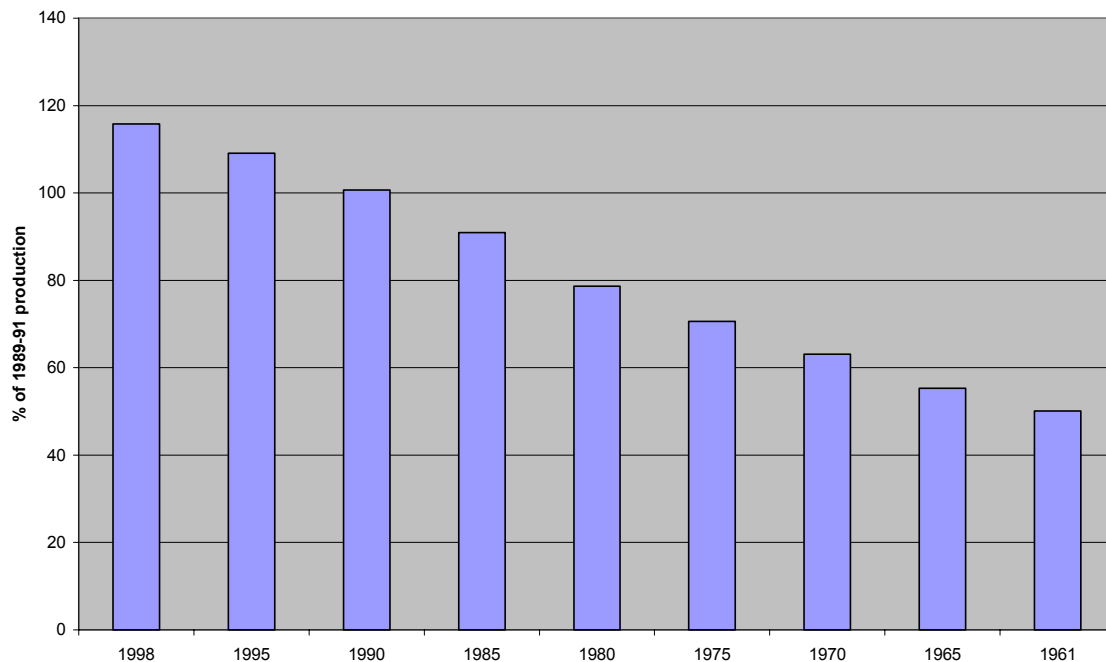
7) Nutrient Management: Proper management of nitrogen and other plant nutrients can improve the soil and protect the environment. Increased use of on-farm nutrient sources, such as manure and leguminous cover crops, also reduces purchased fertilizer costs.

8) Agroforestry: Trees and other woody perennials are often underutilized on farms and ranches. Agroforestry covers a range of tree uses on farms, including interplanting trees (such as walnuts) with crops or pasture, better managing woodlots, and using trees and shrubs along streams as riparian buffer strips.

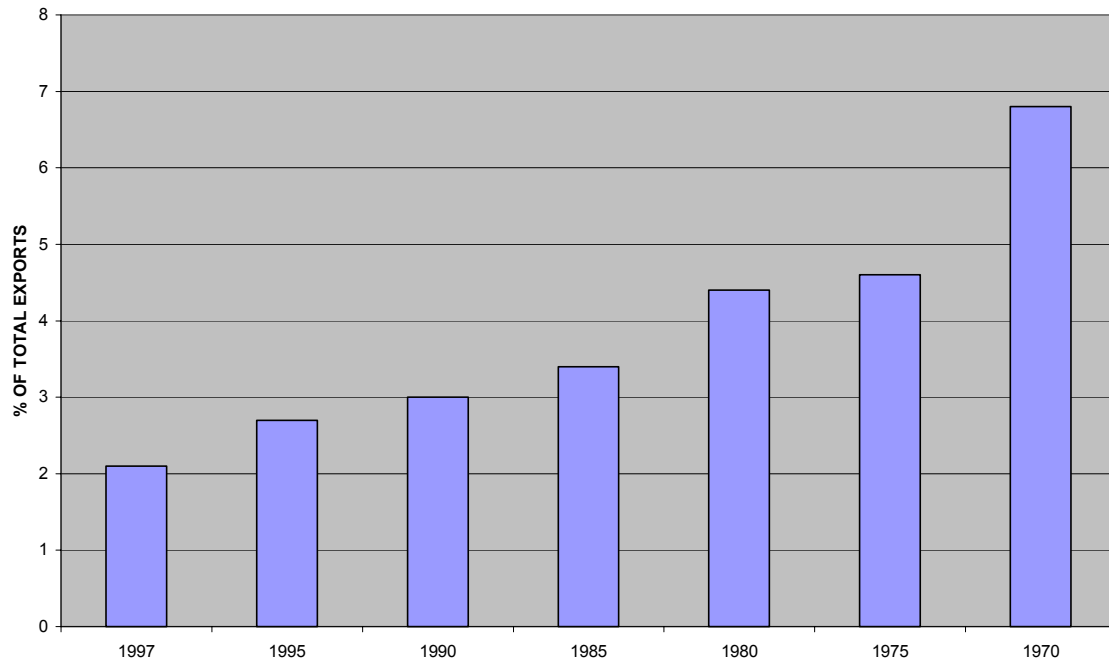
9) Marketing: Farmers and ranchers across the country are finding that improved marketing provides a key way to enhance profitability. Direct marketing of agricultural goods to consumers is becoming much more common, including through farmers' markets, roadside stands and community-supported agriculture.

The following graphs show global trends in agricultural production, exports, imports and fertilizer use. Note the relationship between production and fertilizer, and the trends in agricultural exports and imports.

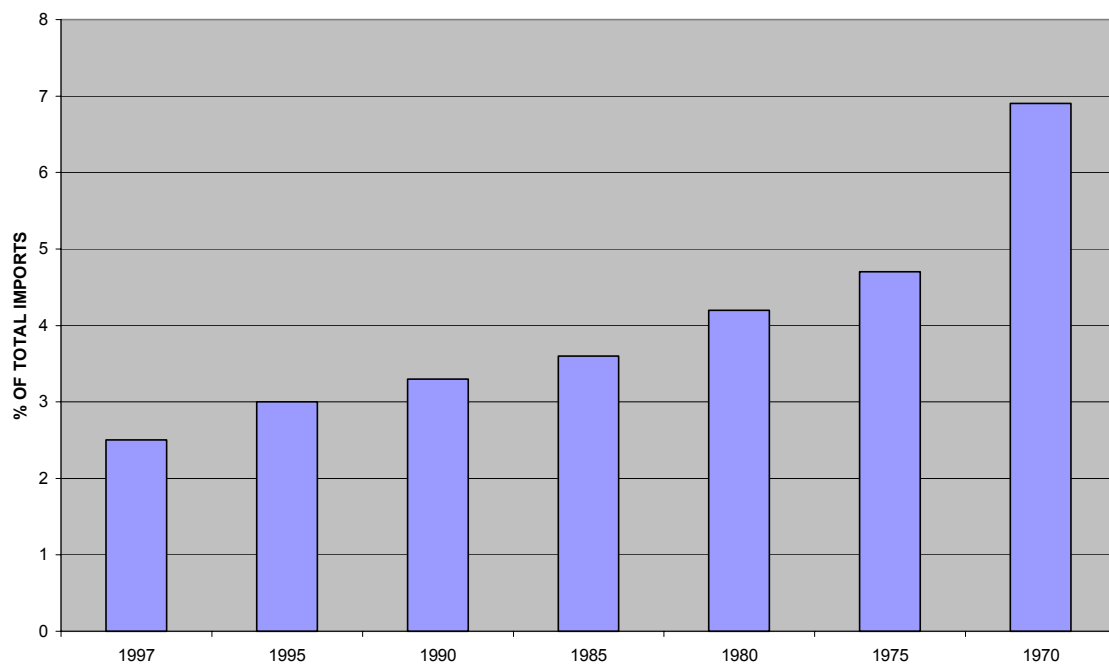
WORLD AGRICULTURAL PRODUCTION



WORLD AGRICULTURAL EXPORTS



WORLD AGRICULTURAL IMPORTS



WORLD FERTILIZER USE

